De novo design of bispecific proteins

Proposed workflow | Miniprotein binders, VHHs and antibodies

A DISCOVER EACH ARM

● Arm A ● Arm B Repeat per target

1 Define epitopes

Structure · accessibility
Native target context

2 Generate arms

RFdiffusion → ProteinMPNN
or BindCraft
RFantibody (antibody route)

3 Reassess complexes

AlphaFold2-Multimer / AF3
Fold · interface · epitope

4 Select + test arms

Diversity · developability
Binding + specificity

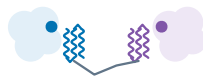
B ENGINEER AND VALIDATE

Selected arms A + B

5 Combine arms

Linker · order · valency
IgG: chain pairing

6 Assess construct

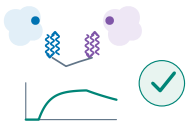


AlphaFold2 / AF3
Clashes · target access

7 Produce + QC

Expression · purification
SEC–MALS · stability

8 Test dual function



SPR / BLI · simultaneous binding
Specificity · cellular function

Assay-guided redesign

Conceptual schematics; curves illustrate assays, not measured results.

Green: experimental assessment